

AMARDEEP SAHA

Software Engineer

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LinkedIn | GitHub | Portfolio | LeetCode

Summary

Backend Software Engineer with 3+ years of experience designing and scaling distributed backend systems using Golang, Kafka, PostgreSQL, and Redis. Built a concurrent caching layer that cut database calls by over 90% and Kafka-driven pipelines for real-time data validation and enrichment, alongside LLM-orchestration pipelines (LangGraph, LangChain) for intelligent, cost-optimized data processing. Strong foundation in data structures, algorithms, and system design, with a track record of improving latency, throughput, and conversion metrics in production.

Skills

Programming Languages: Golang, Python, JavaScript, TypeScript, SQL, Solidity, C++, C, PHP, Bash, HTML

Backend & Frameworks: Gin, Node.js, Express.js, REST APIs, Apache Kafka, Redux, Next.js, React

AI / LLM Tooling: LangChain, LangGraph, Langfuse, HuggingFace Transformers, Groq SDK, Prompt Engineering

Databases & Caching: PostgreSQL, Cassandra, Redis, MongoDB, SQLite

Tools & Platforms: Docker, Kubernetes, Git, Windmill, Devtron, Remix, Ethereum

Core Competencies and Methodologies: Distributed Systems, System Design, Concurrency Control, Distributed Locking, CDC, Microservices, Caching Strategies, Data Structures & Algorithms, Performance Optimization

Experience

Indiamart

Noida, India

Backend Software Engineer | June 2023 – Present

- Built a distributed Intent-to-Lead pipeline processing **2.15M+ buyer clickstream events/day**, using **Cassandra** for high-throughput activity logging, scheduled ETL for extracting buyer search interests and enriching product/seller metadata, and **Kafka**-based asynchronous processing for downstream lead creation; converted **~14%** of qualified intents into sold leads.
- Designed and implemented a WhatsApp-based enquiry generation pipeline that identified user intent for specific modules and triggered automated engagement, resulting in **INR 1 lakh additional profit**, scaled onboarding by enabling **200+ additional users per day**.
- Architected and implemented a high-performance concurrent caching layer in **Golang** leveraging in-memory storage and **Redis**, reducing database calls by **90%+** and achieving **~1ms p95 latency** with a **3-second warm-up** and near-zero-latency steady-state reads.
- Led end-to-end full-stack performance optimization initiatives by reviewing **5+ service SRS documents**, identifying lightweight backend API strategies, and transforming sequential service interactions into asynchronous workflows, resulting in a **25% reduction** in overall page load time.
- Designed and developed a **Kafka**-driven location validation and enrichment pipeline in **Golang** with **PostgreSQL**, detecting location mismatches exceeding **250 km** and triggering asynchronous WhatsApp-based remediation workflows to improve data accuracy and automation.
- Implemented Server-Side Rendering (**SSR**) using **React** and **Node.js** to enhance SEO, improve search engine indexability, and drive a **5% increase** in organic traffic.
- Engineered scalable, data-intensive frontend architectures capable of handling **200+ concurrent asynchronous requests** per component while maintaining UI responsiveness, stability, and performance under high load.
- Owned and optimized automated data enrichment workflows for mobile user enquiries, increasing valid form fill rates by **10%** through intelligent extraction and enrichment strategies.
- Architected a unified **JavaScript** versioning and deployment framework that eliminated cross-team release dependencies and streamlined the rollout of centralized shared components.
- Designed and delivered a reusable UI component library comprising **20+ standardized components**, improving design consistency and accelerating feature development across multiple projects.
- Collaborated proactively with cross-functional product, backend, and platform teams to drive rapid iteration cycles, continuous improvements, and efficient delivery of high-impact features.
- Spearheaded the implementation of Google's **one-click unsubscribe** compliance feature for promotional emails, coordinating execution across multiple teams to ensure seamless adoption and regulatory alignment.
- Integrated a sentiment analysis capability using HuggingFace's **multilingual BERT** model to evaluate user tone and intent, enabling smarter downstream processing and enhanced user interaction analysis.
- Designed and orchestrated custom LLM nodes for semantic lead-matching using **LangGraph** and **LangChain**, with **Langfuse** for observability; combined prompt compression, request batching, and flex processing to cut per-request inference cost from **INR 0.80 to under INR 0.10**, sustaining **10K+ requests/day** with response latency **under 10s**.
- Built a concurrency-safe lead-routing pipeline that curates data from **10+ APIs** and routes qualified sellers to users over WhatsApp, using **Redis** distributed locks and atomic **INCR** transactions for race-free processing across distributed goroutines; polled curation status with a **5s** network cap against a **60s SLA**, returning results as soon as all source APIs resolved (**~92% reduction** in worst-case network hold).

QuantumLeap

Remote, India

Research and Development Intern | March 2022 – August 2022

- Engineered a full-stack decentralized application (DAO) supporting **10,000+** members and on-chain transactions, covering governance voting and stake distribution logic.
- Optimized gas-efficient smart contracts in **Solidity** for token distribution and stake allocation, reducing average transaction gas cost by **~18%** through storage layout and loop optimization.
- Implemented frontend-to-blockchain connectivity using **Ethers.js** and **MetaMask**, enabling wallet-based authentication and real-time transaction signing across the platform's **10,000+** member base.
- Deployed and tested smart contracts on the **Ethereum** blockchain using **Remix IDE**, validating contract logic across **20+** test scenarios prior to deployment.

Projects

Voice-Enabled Task Tracker | Next.js, TypeScript, Node.js, Express, SQLite, LLMs, Docker

- Built a backend-driven task management system with REST APIs supporting CRUD operations, filtering, and search using **Node.js**, **Express**, and **SQLite**; the system is capable of handling **500+** concurrent task records with sub-**100ms** query response time, validated in local development testing.
- Designed an LLM-powered natural language processing pipeline using **Groq SDK** to convert voice input into structured task data (title, priority, due date, status); the pipeline is capable of completing end-to-end voice-to-task conversion in under **2 seconds**, based on dev-environment testing.
- Integrated speech-to-text ingestion with backend services, enabling real-time voice-based task creation and Kanban workflows; the system is capable of supporting **10+** concurrent users with real-time board state synchronization in development testing.

Video Conferencing App | *React JS, WebRTC , SocketIO, SQL , NodeJS, Redis, Docker*

- Built a backend-driven task management system with REST APIs supporting CRUD, filtering, and search, using Node.js, Express, and SQLite with clear service separation.
- Designed an LLM-powered natural language processing pipeline using Groq SDK, converting voice input into structured task data (title, priority, due date, status).
- Integrated speech-to-text ingestion with backend services and a Next.js frontend, enabling real-time voice-based task creation and Kanban workflows.

Research Publication: “A Survey Paper on Developments in Blockchain”

- Authored and published a research paper at IIIT Ranchi, analyzing over **25** papers on the evolution of blockchain consensus mechanisms.

Education

Indian Institute of Information Technology, Ranchi

Ranchi, India

B.Tech, Computer Science and Engineering | July 2019 – May 2023 | CGPA: 8.45 / 10